CS

ChainShield

Smart Supply Chain Disruption Detection

AI-powered route optimization for modern logistics

■ Live Demo ■ Global Scale ■ Real-time AI

Google Solution Challenge 2026 · Smart Supply Chains Track

Swipe to see the demo →

THE PROBLEM

Global supply chains are flying blind.

Weather disruptions
detected AFTER delays begin

Route bottlenecks
invisible until cascading failure

Cargo at risk
no proactive alerts for operators

Manual monitoring
too slow for million-shipment scale

ChainShield fixes this. ↓

OUR SOLUTION

ChainShield detects disruptions before they happen.

Real-time Risk Scanning

60-second automated scans across all active shipments

Live Map Dashboard

Interactive global map with shipment positions & alerts

AI Route Optimization

Gemini AI generates instant alternate route solutions

Operator Notifications

Field operators get instant alerts with action steps

ARCHITECTURE

01

Data Layer

OpenWeatherMap · GPS Coords · Historical Routes

02

Risk Engine

4-Factor AI Scoring: Weather 40% · Congestion 25% · Cargo 20% · History 15%

03

Backend API

FastAPI · SQLAlchemy · APScheduler · Firebase

04

Frontend

React 18 · Leaflet Maps · Gemini AI · Role-based Access

DEMO · STEP 1

Secure Login & Role-Based Access

CS

ChainShield

Smart Supply Chain Protection

Sign In | Register

Email Address

admin@chainshield.com

Password

••••••••••

Sign In →

■ Admin ■ Manager ■ Operator

3 user roles — Admin, Manager & Field Operator

Each sees only what they need. JWT + Firebase Auth.

DEMO · STEP 2

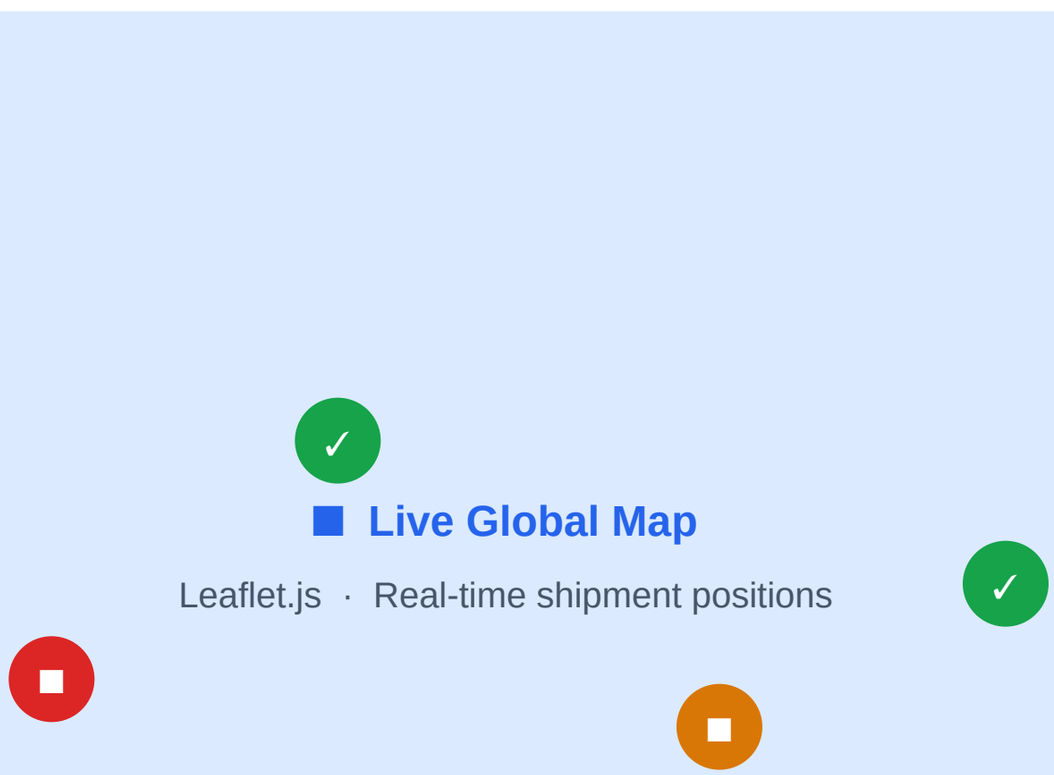
Manager Dashboard:
Full Situational Awareness

Active Disruptions

Rotterdam Port Strike
Critical · 72h delay

Storm — Singapore Route
High · 36h delay

Road Congestion — Mumbai
Medium · 18h delay



3-column layout: disruption list | live map | shipment detail

Real-time data refreshes every 30 seconds automatically.

12	3	87%
Active Shipments	Live Disruptions	On-Time Rate

DEMO · STEP 3

The 4-Factor Risk Engine

Every shipment scored 0–100 every 60 seconds

■■ Weather Severity

Live data from OpenWeatherMap API

40 pts

■ Route Congestion

Deterministic per shipment ID

25 pts

■ Cargo Sensitivity

Pharma, Hazmat, Electronics = High Risk

20 pts

■ Historic Delay Prob.

Country-level delay database

15 pts

< 30 = Low **30–49 = At Risk** **50–69 = Disruption** **≥ 70 = Critical**

DEMO · STEP 4

Auto-Detected Disruption
with AI-Generated Solutions

Severe Storm — Rotterdam → Singapore

CRITICAL

Risk Engine Score: 82/100 (Weather 35/40 · Congestion 22/25 · Cargo 20/20 · Historic 5/15)

Wind: 67 knots · Location: 51.9°, 4.1° · Estimated Delay: 72 hours

■ Risk Engine v2 Detected

■ RECOMMENDED

Divert to Alternate Sea Route

+18h · +$14,000 · Risk Score: 8/100

Apply Solution

■■ ALTERNATIVE

Hold Position — Wait Out Storm

+48h · +$5,500 · Risk Score: 35/100

Manager clicks 'Apply' → operator notified instantly.

DEMO · STEP 5

Gemini AI Assistant:

Ask Anything About Your Supply Chain

ChainShield AI Assistant (Powered by Google Gemini)

What's causing the Rotterdam delay?

The Rotterdam disruption (DIS-A4F2C1) was flagged by Risk Engine v2 with a score of 82/100. Primary cause: Severe storm with 67-knot winds. I recommend 'Divert to Alternate Sea Route' — adds 18h but keeps risk at 8/100.

Which shipments are most at risk right now?

3 shipments flagged: SHP-001 (score 82, disrupted), SHP-007 (score 54, at-risk), SHP-012 (score 48, at-risk). SHP-001 carries Pharmaceuticals — priority intervention.

Ask about routes, risks, shipments…

→

DEMO · STEP 6

Field Operator Dashboard:
Shipment Status & Live Alerts

■ Truck Driver

● ACTIVE

Alex Kumar · RoadHaul Co.

Chat #: 47291 · License: LIC-TK-9823

SHP-007: Mumbai → Frankfurt

Cargo: Pharmaceuticals · Mode: Road · Progress: 62%

Risk: 54 / 100

■ **Route Alert: Congestion on NH-48 near Pune**

Solution applied: Bypass via Ring Road. ETA updated.

Extra: +3h · Cost: +$600 · New risk: 7/100

■ **Direct Message from Manager (Chat #12847)**

"Take the ring road bypass — I've applied the alternate route."

Auto-refreshes every 15 seconds. Full offline-resilient design.

TECHNOLOGY STACK

Built with modern, production-grade tools

Frontend	Backend	AI & Data	DevOps
■■ React 18 + Vite	■ FastAPI (Python)	■ Google Gemini API	■ Vercel (Frontend)
■■ Leaflet.js (Maps)	■■ SQLAlchemy + SQLite	■■ OpenWeatherMap API	■■ Render (Backend)
■ Inline CSS + Context API	■■ APScheduler (60s scan)	■ Firebase Firestore	■ Docker + uvicorn
■ Firebase Auth	■ JWT + Firebase Admin	■ 4-Factor Risk Model	■ Git Submodules

Solution Challenge 2026

ChainShield is live & open-source.

12+	4-Factor	60s	3 Roles
Shipments Tracked	AI Risk Scoring	Auto-Scan Interval	Admin / Mgr / Operator

chainshield-eight.vercel.app

API: chainshield.onrender.com

Built by Bhavik Visani · Google Solution Challenge 2026

If this helped you, drop a like or comment below!